

Additional Topic – Ratios of Sums/Differences of Angles

Review

θ Radian	θ degrees	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0°	0	1	0
$\frac{\pi}{6}$	30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4}$	45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$\frac{\pi}{3}$	60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	90°	1	0	ND

Pythagorean Trigonometric Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Key Idea – Ratios of Sums/Differences of Angles

In general, $\sin(A + B) \neq \sin A + \sin B$.

To find the exact value of the sums/differences of two angles the following rules may be used:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

These rules are known as the sum and difference identities.

The identities may be summarised:

$$\begin{aligned}\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}\end{aligned}$$

The exact trigonometric ratio for 30° , 45° and 60° are known. The identities for the sum and differences of angles may be used to find the exact trigonometric ratios for other angles.

Example:

Find the exact value of $\sin 75^\circ$.

$$\begin{aligned}\sin 75^\circ &= \sin(30^\circ + 45^\circ) \\ &= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ \\ &= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} \\ &= \frac{1 + \sqrt{3}}{2\sqrt{2}} \\ &= \frac{\sqrt{2} + \sqrt{6}}{4}\end{aligned}$$

Exercise:

Find the exact value of $\sin 15^\circ$.

$$\sin 15^\circ = \sin(60^\circ - 45^\circ)$$

$$= \sin 60^\circ \cos 45^\circ - \cos 60^\circ \sin 45^\circ$$

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$$

$$= \frac{(\sqrt{3} - 1)}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{6} - \sqrt{2}}{2 \times 2} \longrightarrow = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

Exercise:

Find the exact value of $\tan 105^\circ$.

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\tan 105^\circ = \tan(60^\circ + 45^\circ)$$

$$= \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \times \tan 45^\circ}$$

$$= \frac{\sqrt{3} + 1}{1 - \sqrt{3} \times 1}$$

$$= \frac{(\sqrt{3} + 1) \times (1 + \sqrt{3})}{(1 - \sqrt{3}) \times (1 + \sqrt{3})}$$

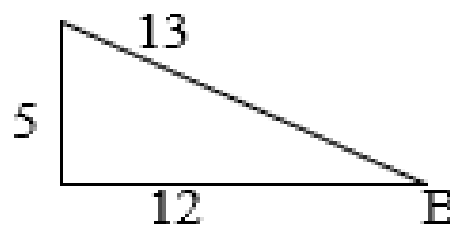
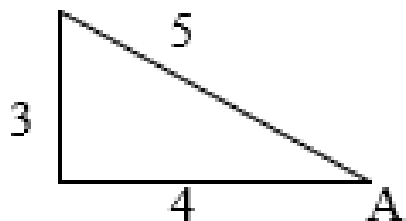
$$= \frac{\sqrt{3} + 3 + 1 + \sqrt{3}}{1 - 3} \longrightarrow = \frac{4 + 2\sqrt{3}}{-2} = -2 - \sqrt{3}$$

In the previous examples, the exact trigonometric ratios of 30° , 45° , 60° were used. If given the exact trigonometric values of other angles, the sum and differences identities may also be used.

Example:

If A and B are acute angles and $\sin A = \frac{3}{5}$, and $\cos B = \frac{12}{13}$ find the exact value of $\sin(A + B)$.

Step 1. Use right angle triangles to find $\cos A$ and $\sin B$:



$$\cos A = \frac{4}{5}$$

$$\sin B = \frac{5}{13}$$

Step 2. Write the formula: $\sin(A + B) = \sin A \cos B + \cos A \sin B$

Step 3. Substitute into the formula:

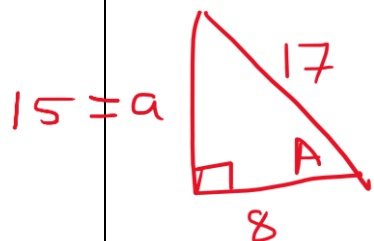
$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$= \frac{3}{5} \times \frac{12}{13} + \frac{4}{5} \times \frac{5}{13}$$

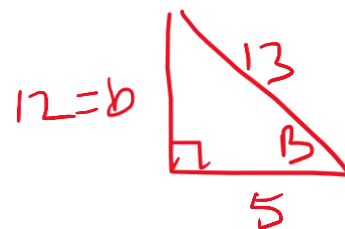
$$= \frac{63}{65}$$

Exercise:

If A and B are acute angles and $\cos A = \frac{8}{17}$, and $\cos B = \frac{5}{13}$ find the exact value of $\cos(A + B)$.



$$\begin{aligned}a^2 + 8^2 &= 17^2 \\a^2 &= 17^2 - 8^2 \\a &= \sqrt{17^2 - 8^2} \\a &= 15\end{aligned}$$



$$\begin{aligned}b^2 + 5^2 &= 13^2 \\b^2 &= 13^2 - 5^2 \\b &= \sqrt{13^2 - 5^2} \\b &= 12\end{aligned}$$

$$\begin{aligned}\cos(A+B) &= \cos A \cos B - \sin A \sin B \\&= \frac{8}{17} \times \frac{5}{13} - \frac{15}{17} \times \frac{12}{13} \\&= \frac{40}{221} - \frac{180}{221} \\&= \frac{-140}{221}\end{aligned}$$

Other relationships may be derived from the sum and difference identities.

Example:

Show that $\tan(45^\circ + x) \equiv \frac{1 + \tan x}{1 - \tan x}$

$$\mathbf{LHS} = \tan(45^\circ + x)$$

$$= \frac{\tan 45^\circ + \tan x}{1 - \tan 45^\circ \tan x}$$

$$= \frac{1 + \tan x}{1 - \tan x}$$

$$= \mathbf{RHS}$$

Exercise:

Show that $\sin(45^\circ + x) \equiv \frac{1}{\sqrt{2}}(\cos x + \sin x)$.

$$\begin{aligned}\text{LHS} &= \sin(45^\circ + x) \\ &= \sin 45^\circ \cos x + \cos 45^\circ \sin x \\ &= \frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \\ &= \frac{1}{\sqrt{2}} (\cos x + \sin x) \\ &= \text{RHS}\end{aligned}$$

When simplifying expressions involving products of trigonometric ratios, compare the question with the identities for the sum or difference of angles.

Example: Evaluate $\cos 70^\circ \cos 10^\circ + \sin 70^\circ \sin 10^\circ$

$$\begin{aligned}\cos 70^\circ \cos 10^\circ + \sin 70^\circ \sin 10^\circ &= \cos(70 - 10)^\circ \\ &= \cos 60^\circ \\ &= 0.5\end{aligned}$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

Exercise:

Use sum and difference identities to evaluate the following:

(a) $\sin 10^\circ \cos 20^\circ + \cos 10^\circ \sin 20^\circ = \sin(10^\circ + 20^\circ)$
 $= \sin 30^\circ$
 $= \frac{1}{2}$

(b) $\frac{\tan 20^\circ + \tan 25^\circ}{1 - \tan 20^\circ \tan 25^\circ} = \tan(20^\circ + 25^\circ)$
 $= \tan 45^\circ$
 $= 1$

Class Questions:

Ratios of Sums/Differences of Angles

1. Write the expansion for the following:

(i) $\sin(\alpha + \beta)$ (ii) $\tan(2x - y)$ (iii) $\sin(3A + B) + \sin(3A - B)$

2. Simplify the following:

(i) $\cos \alpha \cos \beta + \sin \alpha \sin \beta$ (ii) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$

(iii) $\frac{\tan 3A + \tan 2B}{1 - \tan 3A \tan 2B}$

3. Find the exact value of the following:

(i) $\cos 15^\circ$ (ii) $\tan 75^\circ$

(iii) $\sin 75^\circ$ (iv) $\tan 15^\circ$

4. If $\sin \alpha = \frac{4}{5}$, $\sin \beta = \frac{5}{13}$ and α and β are acute find the exact value of:

(i) $\sin(\alpha + \beta)$ (ii) $\cos(\alpha + \beta)$

(iii) $\cos(\alpha - \beta)$ (iv) $\sin(\beta + \beta)$

5. If A and B are acute angles and $\sin A = \frac{3}{5}$, $0 < A < \frac{\pi}{2}$, and $\sin B = \frac{1}{2}$, $0 < B < \frac{\pi}{2}$,

find the exact value of :

(i) $\sin(A + B)$. (ii) $\cos(A - B)$

6. Prove the following identities:

(i) $\cos(A + B)\cos B + \sin(A + B)\sin B \equiv \cos A$

(ii) $\frac{\cos(A + B)}{\sin A \sin B} \equiv \cot A \cot B - 1$

(iii) $\frac{\sin(A - B)}{\sin A \sin B} \equiv (\cot B - \cot A)$.

Solutions to Class Questions:

Ratios of Sums/Differences of Angles

1. (i) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$

(ii) $\frac{\tan 2x - \tan y}{1 + \tan 2x \tan y}$

(iii) $\sin 3A \cos B + \cos 3A \sin B + \sin 3A \cos B - \cos 3A \sin B$
 $= 2 \sin 3A \cos B$

2. (i) $\cos(\alpha - \beta)$

(ii) $\sin(\alpha + \beta)$

(iii) $\tan(3A + 2B)$

3.

(i)

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ \cos 15^\circ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}} \\ &= \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

(iii)

$$\begin{aligned}\sin 75^\circ &= \sin(45^\circ + 30^\circ) \\ \sin 75^\circ &= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}} \\ &= \frac{\sqrt{6}+\sqrt{2}}{4}\end{aligned}$$

(ii)

$$\begin{aligned}\tan 75^\circ &= \tan(45^\circ + 30^\circ) \\ \tan 75^\circ &= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} \\ &= \frac{1 + \frac{1}{\sqrt{3}}}{1 - 1 \cdot \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} \\ &= \frac{4+2\sqrt{3}}{2} = 2 + \sqrt{3}\end{aligned}$$

(iv)

$$\begin{aligned}\tan 15^\circ &= \tan(45^\circ - 30^\circ) \\ \tan 15^\circ &= \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} \\ &= \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \cdot \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{\sqrt{3}-1}{\sqrt{3}+1} \\ &= \frac{4-2\sqrt{3}}{2} = 2 - \sqrt{3}\end{aligned}$$

4. If $\sin \alpha = \frac{4}{5}$, $0 < \alpha < \frac{\pi}{2}$ then adjacent side = 3

If $\sin \beta = \frac{5}{13}$, $0 < \beta < \frac{\pi}{2}$ then adjacent side = 12

$$\begin{aligned}\text{(i) } \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \sin \beta \cos \alpha & \text{(ii) } \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \frac{4}{5} \cdot \frac{12}{13} + \frac{5}{13} \cdot \frac{3}{5} & &= \frac{3}{5} \cdot \frac{12}{13} - \frac{4}{5} \cdot \frac{5}{13} \\ &= \frac{63}{65} & &= \frac{16}{65}\end{aligned}$$

$$\begin{aligned}\text{(iii) } \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta & \text{(iv) } \sin(\beta + \beta) &= \sin \beta \cos \beta + \cos \beta \sin \beta \\ &= \frac{3}{5} \cdot \frac{12}{13} + \frac{4}{5} \cdot \frac{5}{13} & &= \frac{5}{13} \cdot \frac{12}{13} + \frac{12}{13} \cdot \frac{5}{13} \\ &= \frac{56}{65} & &= \frac{120}{169}\end{aligned}$$

5. If $\sin A = \frac{3}{5}$ then $\cos A = \frac{4}{5}$, if $\sin B = \frac{1}{2}$ then $\cos B = \frac{\sqrt{3}}{2}$

$$\begin{aligned}\text{(i) } \sin(A + B) &= \sin A \cos B + \cos A \sin B & \text{(ii) } \cos(A - B) &= \cos A \cos B + \sin A \sin B \\ &= \frac{3}{5} \cdot \frac{\sqrt{3}}{2} + \frac{4}{5} \cdot \frac{1}{2} & &= \frac{4}{5} \cdot \frac{\sqrt{3}}{2} + \frac{3}{5} \cdot \frac{1}{2} \\ &= \frac{3\sqrt{3}+4}{10} & &= \frac{4\sqrt{3}+3}{10}\end{aligned}$$

$$5. \quad (i) \quad \cos(A+B)\cos B + \sin(A+B)\sin B \equiv \cos A$$

$$LHS = \cos((A+B)-B)$$

$$= \cos A$$

$$= RHS$$

(ii)

$$\frac{\cos(A+B)}{\sin A \sin B} \equiv \cot A \cot B - 1$$

$$LHS = \frac{\cos A \cos B - \sin A \sin B}{\sin A \sin B}$$

$$= \frac{\cos A \cos B}{\sin A \sin B} - \frac{\sin A \sin B}{\sin A \sin B}$$

$$= \frac{\cos A}{\sin A} \cdot \frac{\cos B}{\sin B} - 1$$

$$= \cot A \cot B - 1$$

$$= RHS$$

(iii)

$$\frac{\sin(A-B)}{\sin A \cos B} \equiv (\cot B - \cot A)$$

$$LHS = \frac{\sin(A-B)}{\sin A \sin B}$$

$$= \frac{\sin A \cos B - \sin B \cos A}{\sin A \sin B}$$

$$= \frac{\sin A \cos B}{\sin A \sin B} - \frac{\sin B \cos A}{\sin A \sin B}$$

$$= \frac{\cos B}{\sin B} - \frac{\cos A}{\sin A}$$

$$= \cot B - \cot A$$

$$= RHS$$